JINGNAN DU

Department of Psychology, University of Notre Dame
Website: https://brainmaplab.nd.edu/
Email: jdu6@nd.edu **Phone:** 617-803-6628
Office: E362 Corbett Family Hall, Notre Dame, IN 46556

ACADEMIC POSITIONS

University of Notre Dame, Department of Psychology 2026 - present
Assistant Professor

Harvard University, Department of Psychology 2021 - 2026
Postdoctoral Researcher (Buckner Lab)

MIT, Department of Brain and Cognitive Sciences 2025 - 2026
Visiting Postdoc (Fedorenko lab)

DEGREES AND EDUCATION

Harvard University 2019 - 2021
Visiting Ph.D. in Psychology

Fudan University 2021
Ph.D. in Applied Mathematics

Fudan University 2018
M.S. in Mathematics

Xidian University 2015
B.S. in Applied Mathematics

PUBLICATIONS

1. **Du J**, Billot A, Sun W, Hickok G, Eldaief MC, Buckner RL. "Multiple nested distributed language networks in the human brain." *bioRxiv* 2026. [pdf]
2. Zhi D, **Du J**, Whitfield-Gabrieli S, Diedrichsen J, Ge T. "Precision functional parcellation of the human cortex via rest-task fMRI fusion." *bioRxiv* 2026. [pdf]
3. Elliott M, **Du J**, Nielsen JA, Hanford LC, Kivisäkk P, Arnold SE, Dickerson BC, Mair RW, Eldaief MC, Buckner RL. "Precision estimates of longitudinal brain aging capture unexpected individual differences in one year." *Nature Communications* 17:2401, 2026. [pdf]
4. Hanford LC, Eshghi M, **Du J**, Buckner RL, Mair RW, Ge T, Juttukonda MR, Salat DH. "Age-associated structural decline is linked to arterial flow territories in the brain: Insights from lifespan human connectome project in aging." *medRxiv* 2026. [pdf]
5. Lynch CJ, Chang M, Elbau I, Gordon EM, Laumann TO, **Du J**, Ladwig Z, et al. "Objective Quality Assessment for Precision Functional MRI Data." *Neuron* 2026. [pdf]
6. Libowitz MR, Sun W, Rabinovich R, **Du J**, Campbell JM, Cowan RL, Shahdoust N, Price TA, Davis TS, Buckner RL, et al. "Increased aperiodic exponents track depressive symptom severity in refractory epilepsy." *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging* 2026. [pdf]
7. **Du J**, Tripathi V, Elliott ML, Ladopoulou J, Sun W, Eldaief MC, Buckner RL. "Within-individual precision mapping of brain networks exclusively using task data." *Neuron* 113:4069–4083.e6, 2025. [link]

8. Sun W, Billot A, **Du J**, Wei X, Lemley RA, Daneshzand M, Nummenmaa A, Buckner RL, Eldaief MC. "Precision network modeling of TMS across individuals suggests therapeutic targets and potential for improvement." *Human Brain Mapping* 46:e70266, 2025. [pdf]
9. Kosakowski HL, **Du J**, Tripathi V, Eldaief MC, Buckner RL. "Ventral striatum is preferentially correlated with the salience network including regions in dorsolateral prefrontal cortex." *Journal of Neurophysiology* 134:193-215, 2025. [pdf]
10. Saadon-Grosman N, **Du J**, Kosakowski HL, Angeli PA, DiNicola LM, Eldaief MC, Buckner RL. "Within-individual organization of the human cognitive cerebellum: Evidence for closely juxtaposed, functionally specialized regions." *Science Advances* 10:eadq4037, 2024. [pdf]
11. **Du J**, DiNicola LM, Angeli PA, Saadon-Grosman N, Sun W, Kaiser S, Ladopoulou J, Xue A, Yeo BTT, Eldaief MC, Buckner RL. "Organization of the human cerebral cortex estimated within individuals: Networks, global topography, and function." *Journal of Neurophysiology* 131:1014-82, 2024. [pdf]
12. Kosakowski HL, Saadon-Grosman N, **Du J**, Eldaief MC, Buckner RL. "Human striatal association megaclusters." *Journal of Neurophysiology* 131:1083-100, 2024. [pdf]
13. Gong W, Fu Y, Wu B-S, **Du J**, Yang L, Zhang YR, Chen SD, Kang J, Mao Y, Dong Q, Tan L, Feng J, Cheng W, Yu J-T. "Whole-exome sequencing identifies protein-coding variants associated with brain iron in 29,828 individuals." *Nature Communications* 15:5540, 2024. [pdf](co-first author)
14. **Du J**, Rolls ET, Gong W, Cao M, Vatansever D, Zhang J, Kang J, Cheng W, Feng J. "Association between parental age, brain structure, and behavioral and cognitive problems in children." *Molecular Psychiatry* 27:967-75, 2022. [pdf]
15. Wang L, Zhou C, Cheng W, Rolls ET, Huang P, Ma N, Liu Y, Zhang Y, Guan X, Guo T, et al. "Dopamine depletion and subcortical dysfunction disrupt cortical synchronization and metastability affecting cognitive function in Parkinson's disease." *Human Brain Mapping* 43:1598-610, 2022. [pdf]
16. Yang A, Rolls ET, Dong G, **Du J**, Li Y, Feng J, Cheng W, Zhao XM. "Longer screen time utilization is associated with the polygenic risk for Attention-deficit/hyperactivity disorder with mediation by brain white matter microstructure." *EBioMedicine* 80:104039, 2022. [pdf]
17. **Du J** and Buckner RL. "Precision estimates of macroscale network organization in the human and their relation to anatomical connectivity in the marmoset monkey." *Current Opinion in Behavioral Sciences* 40:144-52, 2021. [pdf]
18. **Du J**, Palaniyappan L, Liu Z, Cheng W, Gong W, Zhu M, Wang J, Zhang J, Feng J. "The genetic determinants of language network dysconnectivity in drug-naïve early stage schizophrenia." *npj Schizophrenia* 7:18, 2021. [pdf]
19. Gong W, Rolls ET, **Du J**, Cheng W, Feng J. "Brain structure is linked to the association between family environment and behavioral problems in children in the ABCD study." *Nature Communications* 12:3769, 2021.[pdf](co-first author)
20. **Du J**, Liu Z, Hanford LC, Anderson KM, Feng J, Ge T, Buckner RL. "Exploration of Alzheimer's disease MRI biomarkers using APOE4 carrier status in the UK Biobank." *medRxiv* 2021. [pdf]
21. Cheng W, Rolls ET, Gong W, **Du J**, Zhang J, Zhang XY, Li F, Feng J. "Sleep duration, brain structure, and psychiatric and cognitive problems in children." *Molecular Psychiatry* 26:3992-4003, 2021. [pdf]
22. **Du J,** Rolls ET, Cheng W, Li Y, Gong W, Qiu J, Feng J. "Functional connectivity of the orbitofrontal cortex, anterior cingulate cortex, and inferior frontal gyrus in humans." *Cortex* 123:185-99, 2020. [pdf]
23. Rolls ET, Cheng W, **Du J**, Wei D, Qiu J, Dai D, Zhou Q, Xie P, Feng J. "Functional connectivity of the right inferior frontal gyrus and orbitofrontal cortex in depression." *Social Cognitive and Affective Neuro-*

science 15:75-86, 2020. [pdf](co-first author)

24. Wang H, Rolls ET, Du X, **Du J**, Yang D, Li J, Li F, Cheng W, Feng J. "Severe nausea and vomiting in pregnancy: psychiatric and cognitive problems and brain structure in children." *BMC Medicine* 18:228, 2020. [pdf]
25. Wang L, Cheng W, Rolls ET, Dai F, Gong W, **Du J**, Zhang W, Wang S, Liu F, Wang J, Brown P, et al. "Association of specific biotypes in patients with Parkinson disease and disease progression." *Neurology* 95:e1445-60, 2020. [pdf]
26. Shen C, Luo Q, Chamberlain S, Morgan S, Romero R, **Du J**, Zhao X, Touchette É, Montplaisir J, Vitaro F, Boivin M, et al. "What is the link between Attention-Deficit/Hyperactivity Disorder and sleep disturbance? A multimodal examination of longitudinal relationships and brain structure using large-scale population-based cohorts." *Biological Psychiatry* 88:459-69, 2020. [pdf]
27. Cheng W, Rolls ET, Robbins TW, Gong W, Liu Z, Lv W, **Du J**, Wen H, Ma L, Quinlan EB, Garavan H, Artiges E, et al. "Decreased brain connectivity in smoking contrasts with increased connectivity in drinking." *eLife* 8:e40765, 2019. [pdf]
28. Liu Z, Rolls ET, Liu Z, Zhang K, Yang M, **Du J**, Gong W, Cheng W, Dai F, Wang H, Ugurbil K, Zhang J, Feng J. "Brain annotation toolbox: Exploring the functional and genetic associations of neuroimaging results." *Bioinformatics* 35:3771-8, 2019. [pdf]

TALKS

1. "Parallel Networks in the Human Brain: Organization and Functional Specialization Revealed Within Individuals", *University of Chicago, Department of Psychology, Cognition Workshop*, Chicago, 2027.3 (scheduled).
2. "Parallel Networks in the Human Brain: Organization and Functional Specialization Revealed Within Individuals", *Harvard Medical School, Athinoula A. Martinos Center, Layer-fMRI Seminar*, Boston (virtual), 2026.11 (scheduled).
3. "Parallel Networks in the Human Brain: Organization and Functional Specialization Revealed Within Individuals", *Massachusetts General Hospital, Center for Precision Psychiatry, Precision Psychiatry Seminar*, Boston (virtual), 2026.10 (scheduled).
4. "Revisiting Human Association Networks Using Precision Functional Mapping", *Harvard University, Department of Psychology, Daniel Schacter's lab*, Cambridge, 2026.5.
5. "Organization of Human Large-Scale Brain Networks Estimated Within Individuals", *OpenTalks #101, OpenScience Community*, Beijing (virtual), 2026.4.
6. "Organization of Human Large-Scale Brain Networks Estimated Within Individuals", *Stanford University, Department of Psychology, Russell Poldrack's lab (one-week academic visit)*, Stanford, 2026.2.
7. "Precision Neuroimaging of the Brain", *Session I: State of the Art: Brain Imaging, MIT Medical Imaging Initiative Symposium, Held at American Academy of Arts and Sciences*, Cambridge, 2026.1. [link]
8. "Organization of Human Large-Scale Brain Networks Estimated Within Individuals", *Monash University, Alex Fornito's Neural Systems and Behaviour Lab*, Melbourne (virtual), 2025.11.
9. "Organization of Human Large-Scale Brain Networks Estimated Within Individuals", *University of Notre Dame, Department of Psychology*, Notre Dame, IN, 2025.11.
10. "Organization of Human Large-Scale Brain Networks Estimated Within Individuals", *Massachusetts Institute of Technology, Ev Fedorenko's Language Lab*, Cambridge, 2025.11.
11. "Organization of Human Large-Scale Brain Networks Estimated Within Individuals", *Peking University,*

School of Psychological and Cognitive Sciences, Beijing (virtual), 2025.11.

12. "Mapping Precision Functional Networks in an Individual's Brain", *Tsinghua University, Department of Psychological and Cognitive Sciences*, Beijing (virtual), 2025.10.
13. "Mapping Precision Functional Networks in an Individual's Brain", *University of Delaware, Cognitive Area Seminar*, Newark (virtual), 2025.10.
14. "Organization of Human Large-Scale Brain Networks Estimated Within Individuals", *Harvard University, Department of Psychology*, Cambridge, 2025.10.
15. "Mapping Precision Functional Networks in an Individual's Brain", *South China Normal University, Multimodal Mind Seminar*, Guangzhou (virtual), 2025.10.
16. "Organization of the Human Cerebral Cortex Estimated Within Individuals", *Harvard University, CBS Neurolunch Seminar*, Cambridge, 2025.10.
17. "Within-individual Precision Mapping of Brain Networks", *Chinese Institute for Brain Research*, Beijing (virtual), 2025.3.
18. "Within-Individual Organization of the Human Cerebral Cortex: Networks, Global Topography, and Function", *Mass General Hospital, Center for Genomic Medicine*, Boston, 2025.2.
19. "Within-Individual Organization of the Human Cerebral Cortex: Networks, Global Topography, and Function", *Stanford University, Department of Linguistics, Cory Shain's lab*, Stanford (virtual), 2024.11.
20. "Estimating Precision Brain Networks Using Task Data: Correspondence to Passive Fixation, Predictive Validity, and New Opportunities", *Harvard University, Department of Psychology*, Cambridge, 2024.10.
21. "Within-Individual Organization of the Human Cerebral Cortex: Networks, Global Topography, and Function", *Beijing Normal University, IDG McGovern Institute / Psychology*, Beijing (virtual), 2024.8.
22. "Within-Individual Organization of the Human Cerebral Cortex: Networks, Global Topography, and Function", *Fudan University, ISTBI*, Shanghai (virtual), 2024.7.
23. "Precision Mapping Using a Novel Multi-Session Hierarchical Bayesian Model", *Harvard University, Center for Brain Science*, Cambridge, 2024.5.
24. "Estimating Cortical Networks Using a Multi-Session Hierarchical Bayesian Model", *Harvard University, Department of Psychology*, Cambridge, 2022.7.
25. "Brain Aging and Alzheimer's disease in the UK Biobank", *Harvard University, Department of Psychology*, Cambridge, 2020.9.
26. "Comparative Network Homology in Human and Marmoset Monkey Brains", *Harvard University, Department of Psychology*, Cambridge, 2020.3.

CONFERENCE PRESENTATIONS

1. **Du J**, Du L, Russell A. "Regional subdivision and multi-resolution parcellation of the DU15NET cortical networks." *Society for Neuroscience, Washington, DC* 2026.
2. Du P, **Du J**, Billot A, Sun W, Yeo BTT, Eldaief MC, Buckner RL. "iGENMap: A novel approach for data-efficient precision functional mapping." *Society for Neuroscience, Washington, DC* 2026.
3. **Du J**. "Not Your Average Brain: Individual-Level fMRI as a Paradigm Shift for Cognitive Neuroscience." *Cognitive Neuroscience Society Annual Meeting, Vancouver* 2026 (Selected Symposium, Chair and Speaker).
4. Sun W, Libowitz MR, Feldman D, **Du J**, Davis T, Rahimpour S, Smith EH, King J, Buckner RL, Shofty B. "Towards a Biomarker for Circuit Dysfunction in Neuropsychiatric Disease: Network Antagonism Measured Within Individuals Using Precision Neuroimaging and Intracranial Electroencephalography." *ASSFN,*

Cleveland 2026.

5. **Du J**, Tripathi V, Elliott ML, Ladopoulou J, Sun W, Eldaief MC, Buckner RL. "Within-individual precision mapping of brain networks using task data and its application in identifying thalamic networks." *Society for Neuroscience, San Diego* 2025.
6. Zhi D, **Du J**, Shahshahani L, Nettekoven C, Pinho AL, Diedrichsen J, Ge T. "Precision Functional Mapping of Individual Human Brain using Rest and Task fMRI Data." *Society for Neuroscience, San Diego* 2025.
7. Libowitz MR, Sun W, Feldman D, **Du J**, Davis T, Rahimpour S, Smith EH, King J, Buckner RL, Shofty B. "Electrophysiology of precision functional networks: Concordant insights from fMRI and iEEG in an oddball paradigm." *Society for Neuroscience, San Diego* 2025.
8. Sun W, Libowitz MR, Feldman D, **Du J**, Billot A, Rodrigues K, Eldaief MC, Davis T, Rahimpour S, Smith EH, King J, Buckner RL, Shofty B. "Automated Precision Functional Network Mapping In Neurosurgical Patients Using A Low-Burden Single MRI Session." *Congress of Neurological Surgeons, Los Angeles* 2025.
9. **Du J**, DiNicola LM, Angeli PA, Saadon-Grosman N, Sun W, Kaiser S, Ladopoulou J, Xue A, Yeo BTT, Eldaief MC, Buckner RL. "DU15NET: A novel 15-network parcellation of the cerebral cortex." *Society for Neuroscience, Chicago* 2024.
10. **Du J**, DiNicola LM, Angeli PA, Saadon-Grosman N, Sun W, Kaiser S, Ladopoulou J, Xue A, Eldaief MC, Buckner RL. "Within-individual organization of the human cerebral cortex revisited." *Organization for Human Brain Mapping Conference, Montreal* 2023.
11. Saadon-Grosman N, **Du J**, Angeli PA, DiNicola LM, Eldaief MC, Buckner RL. "Detailed organization of higher-order networks in the cerebellum estimated within the individual." *Organization for Human Brain Mapping Conference, Montreal* 2023.
12. Hanford L, **Du J**, Saadon-Grosman N, Angeli P, DiNicola LM, Bryce N, Flournoy J, Mair R, Buckner RL, McLaughlin K. "Effects of scan acquisition and smoothing on cortical network parcellations." *Organization for Human Brain Mapping, Montreal* 2023.
13. Buckner RL, Saadon-Grosman N, **Du J**, DiNicola LM, Angeli P, Ladopoulou J, Eldaief MC. "The global organization of the cerebral cortex estimated within the individual." *Organization for Human Brain Mapping Conference, Montreal* 2023.
14. **Du J,** Liu Z, Hanford LC, Anderson KM, Feng J, Ge T, Buckner RL. "Explorations of brain aging in the UK Biobank." *Simons Collaboration on Plasticity and the Aging Brain Conference, NYC* 2022.
15. **Du J,** Rolls ET, Gong W, Cao M, Vatansever D, Zhang J, Kang J, Cheng W, Feng J. "Association between parental age, brain, and psychiatric and cognitive problems in children." *Organization for Human Brain Mapping Conference, Rome* 2019.

ACADEMIC SERVICE

Institutional Service
Member, Undergraduate Studies Committee, Department of Psychology, University of Notre Dame, 2026–present

Ad Hoc Reviewer
Proceedings of the National Academy of Sciences (PNAS)
Nature Biomedical Engineering
Nature Communications
Developmental Cognitive Neuroscience
Neurology
BMC Medicine

Science Bulletin
Patterns
Human Brain Mapping
Translational Psychiatry
Neuropsychopharmacology

PROFESSIONAL MEMBERSHIP

Organization for Human Brain Mapping
Society for Neuroscience
Cognitive Neuroscience Society
Association for Psychological Science

TEACHING

Instructor of Record, PSY/NSBH 43518 Mapping the Human Mind (undergraduate seminar), University of Notre Dame, 2026
Teaching Assistant, Computational Neuroscience, Fudan University, 2019
Teaching Assistant, Linear Algebra, Fudan University, 2017-2018

MENTORING

Mentees:
Gavin Scott, Undergraduate Research Assistant, University of Notre Dame (2026 -)
Sofia Cipollone, Undergraduate Research Assistant, University of Notre Dame (2026 -)
Charles Sander, Senior Thesis Student, University of Notre Dame (2026 -)
Qingyi Li, Visiting Graduate Student, University of Notre Dame (2026 -)
Penghui Du, Visiting Graduate Student, Harvard University (2026 -)
Yen-Wen Chen, OHBM Mentee (2026 -; Current: Postdoctoral Associate, George Washington University)
Natalie Delpratt, OHBM Mentee (2026 -; Current: PhD student, Albert Einstein College of Medicine)
Abbey Russell, Research Assistant, Harvard University (2025 -)
Lin Du, Visiting Graduate Student, Harvard University (2025 -)
Negin Javaheri, OHBM Mentee (2025 - 2026; PhD Student, University of Bremen)
Ella Rowsthorn, OHBM Mentee (2025 - 2026; PhD Student, Monash University)
Stephanie Kaiser, Research Assistant, Harvard University (2022 - 2023)
Joanna Ladopoulou, Research Assistant, Harvard University (2022 - 2024; Current: PhD Student in Clinical Psychology at the University of Southern California)
Wendy Sun, MD-PhD Student, Harvard University (2021 - 2026)
Jennie Li, Visiting Undergraduate Student, Harvard University (2020 - 2021)
PPREP Mentor, Harvard Prospective PhD and RA Event in Psychology (2024 - 2025)

HONORS AND AWARDS

Selected for the NCFDD Faculty Success Program - Spring 2027 cohort, University of Notre Dame, Office of the Provost, 2026
Outstanding Graduate Student, Department of Education of Shanghai, 2021
National Scholarship (USD 4,000), Ministry of Education of China, 2019
Outstanding Student of Fudan University, 2019
Meritorious Winner, Interdisciplinary Contest In Modeling, 2014
Second Prize, National Undergraduate Mathematical Modeling Contest, 2013

NEWS AND PUBLIC OUTREACH

2025: Harvard Gazette, "New way to map the unique brain organization of individuals". [link]
2025: Harvard Center for Brain Science, Featured Publication: "Within-individual precision mapping of brain networks exclusively using task data". [link]
2023: Harvard GSAS News, "Mind Mapper". [link]